

Seakeeping and hull form

SDO/NA/SEAKEEP-011 Rev B

RESTRICTED

4.1 Motion limits

Helicopter launch and recovery shall be possible up to sea state 5 with significant wave height of 4.0 m. Limiting motions at the flight deck reference point are 5 degrees single amplitude roll, 3 degrees pitch, and 2.0 m per second significant vertical velocity. Where passive tank or fin stabilisation is fitted, the criteria shall be met with the stabilisation system operating and the residual case shall be reported separately.

5.2 Active fin selection

Active fin stabilisers shall reduce significant roll amplitude by not less than 70 percent at the design speed in beam seas at the natural roll period. Fin area shall be sized on the roll damping requirement at 18 knots; performance falls with the square of ship speed and the fins shall not be relied upon below 8 knots. Fin housings shall not intrude into a main machinery space or breach a watertight boundary below the damage control deck.

2.3 Length to beam ratio

A length to beam ratio between 7.5 and 8.5 is recommended for patrol and escort platforms in the 2,000 to 3,500 tonne displacement band, balancing resistance in the design speed range against transverse stability margin and deck area for combat systems. Block coefficient in the range 0.44 to 0.50 has proven satisfactory for a design speed of 28 knots.